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Transformer I

Transformer

- A transformer differs from a RNN in that it does not have a recurrent structure and it process tokens in the context window all at once
- Transformer learns the dynamic context information via an "attention" mechanism:
 - Self Attention using scaled dot product
 - Multi-head Attention
 - Positional encoding
- Transformer adopts a (multi-layer) encoder-decoder architecture

Drawback of a Single Context Vector

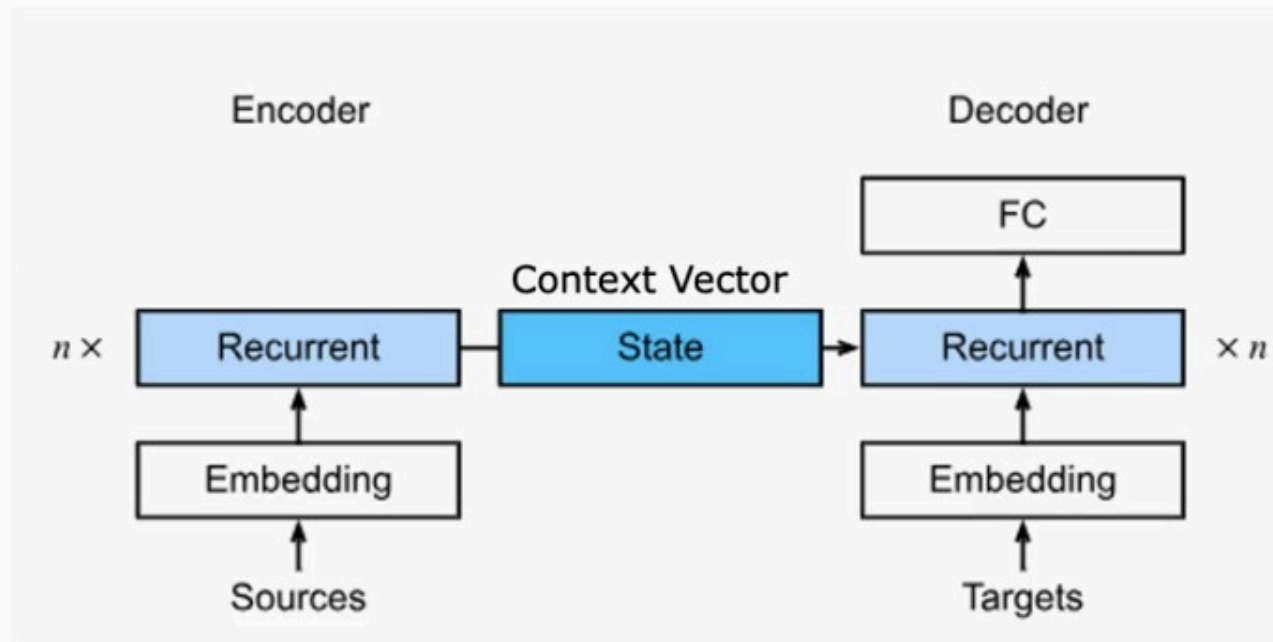
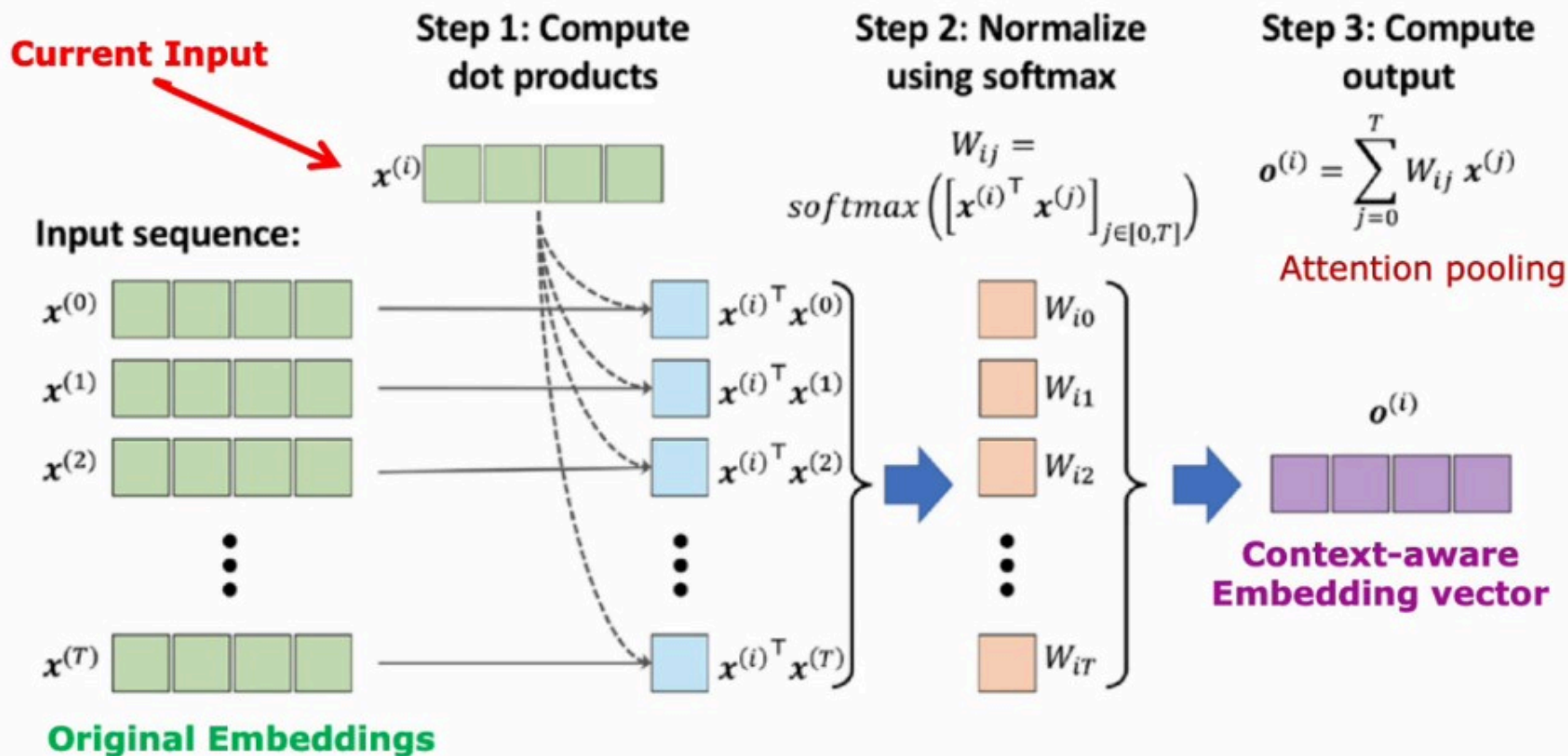


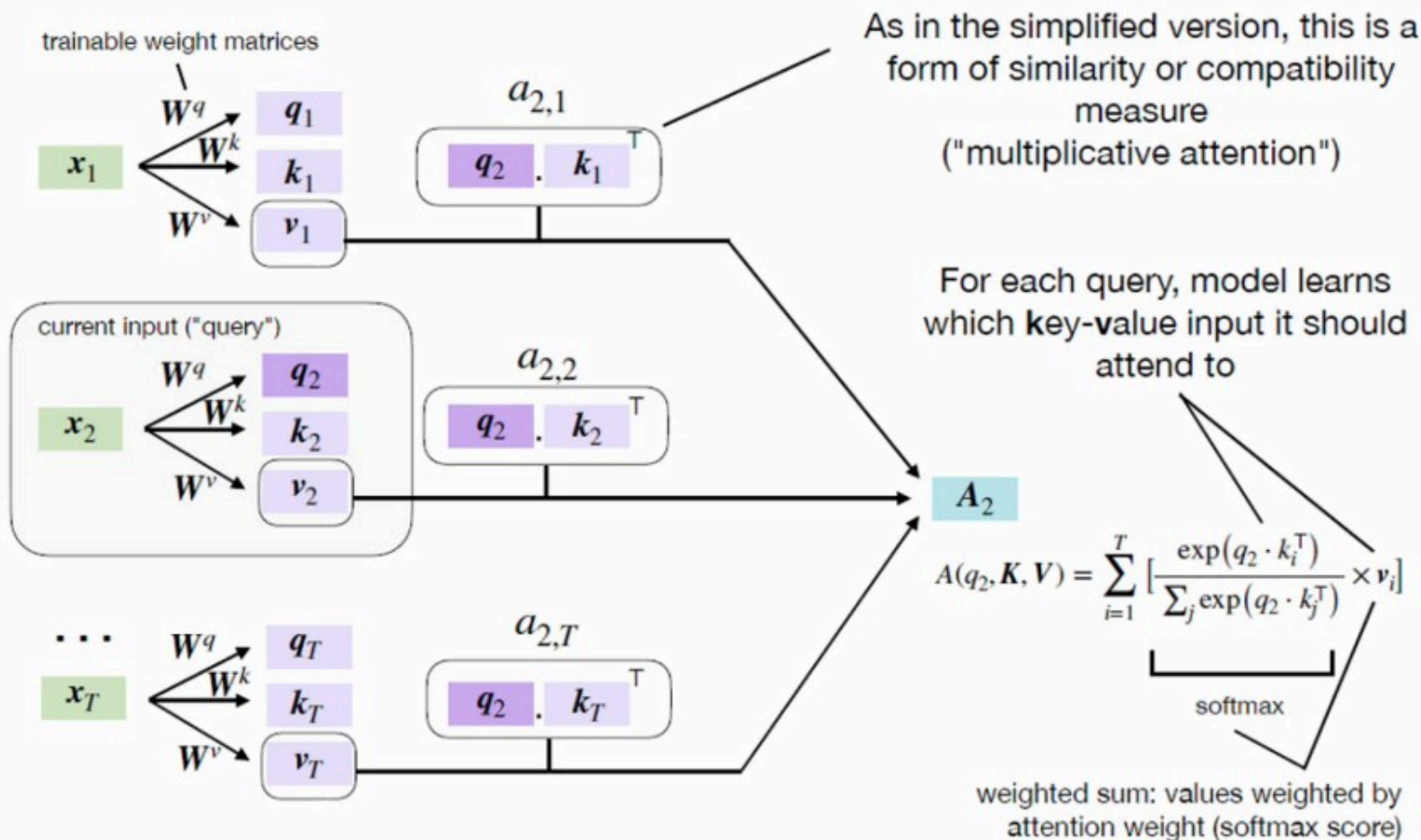
Fig. 11.4.1 Sequence-to-sequence model. The state, as generated by the encoder, is the only piece of information shared between the encoder and the decoder

Not be enough "space" in the intermediate representation to store all that is important in the source sequence $\Rightarrow$ the decoder will fail to translate long and complex sentences.

Self-Attention Mechanism



Self Attention Implementation

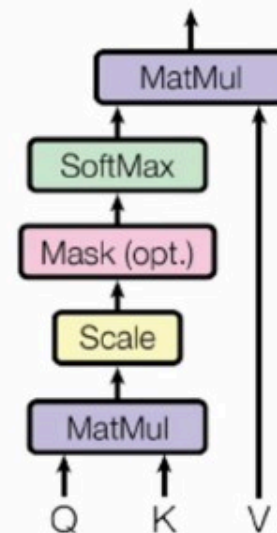


Scaled Product Attention

$$A(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

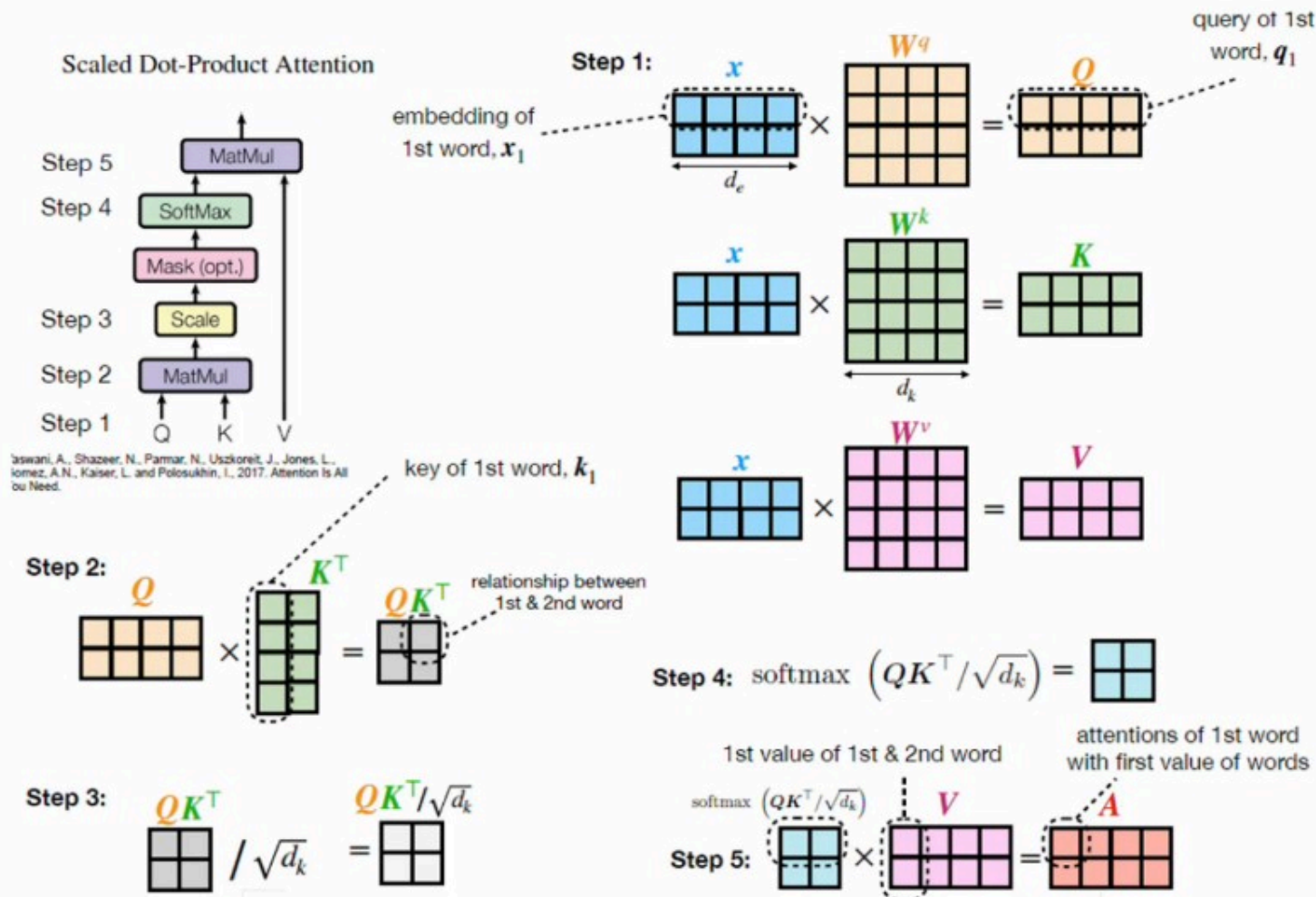
To ensure that the dot-products
between query and key don't
grow too large (and softmax
gradient become too small) for
large d_k

Scaled Dot-Product Attention

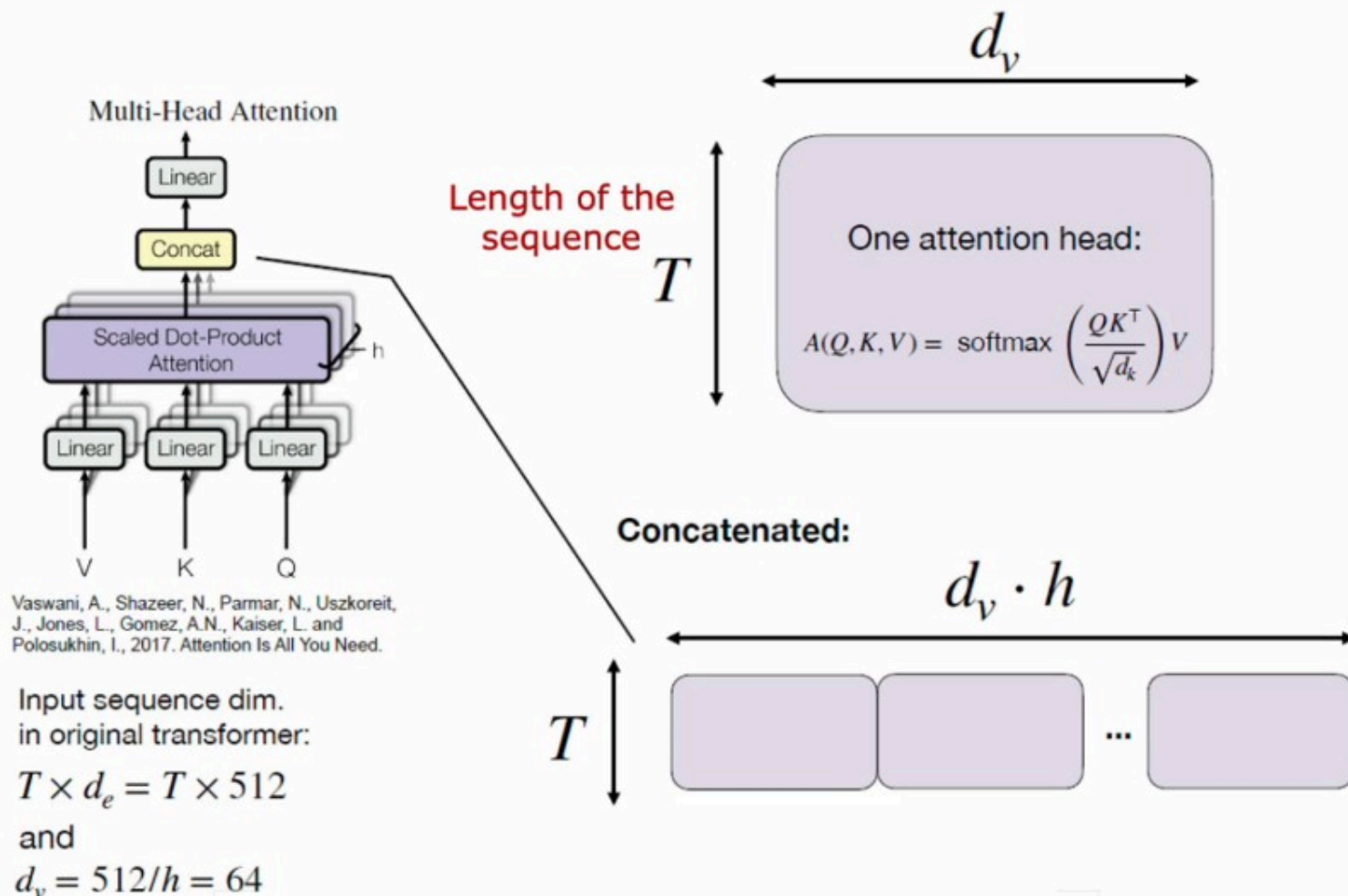


Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L. and Polosukhin, I., 2017. Attention Is All You Need.

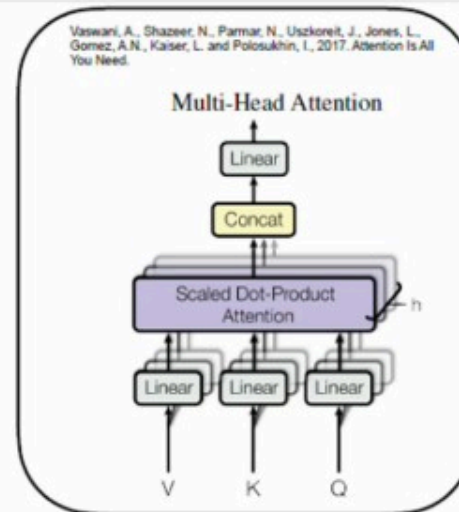
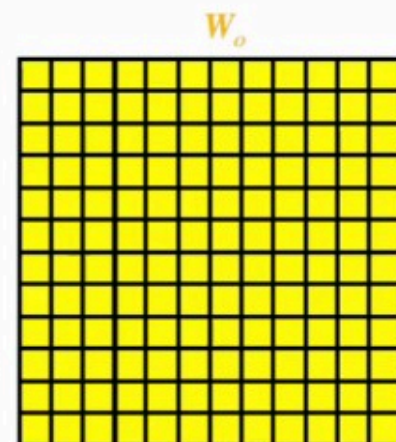
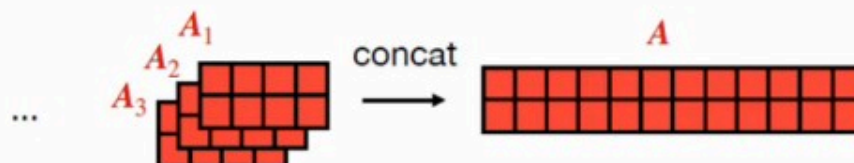
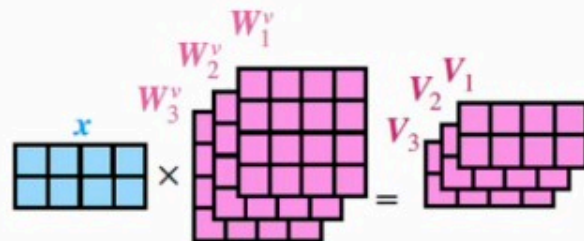
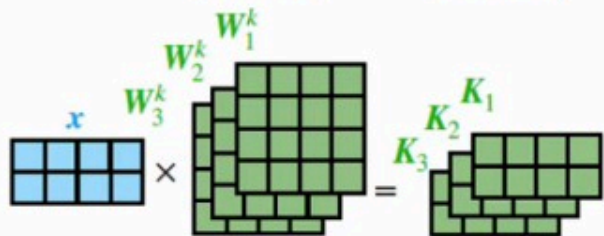
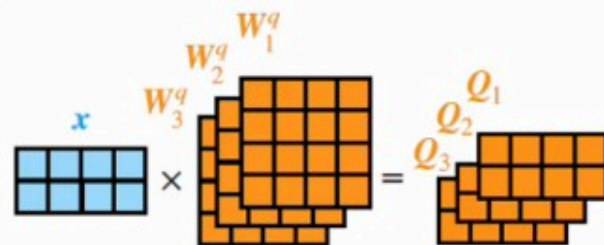
Scaled Dot-Product Implementation



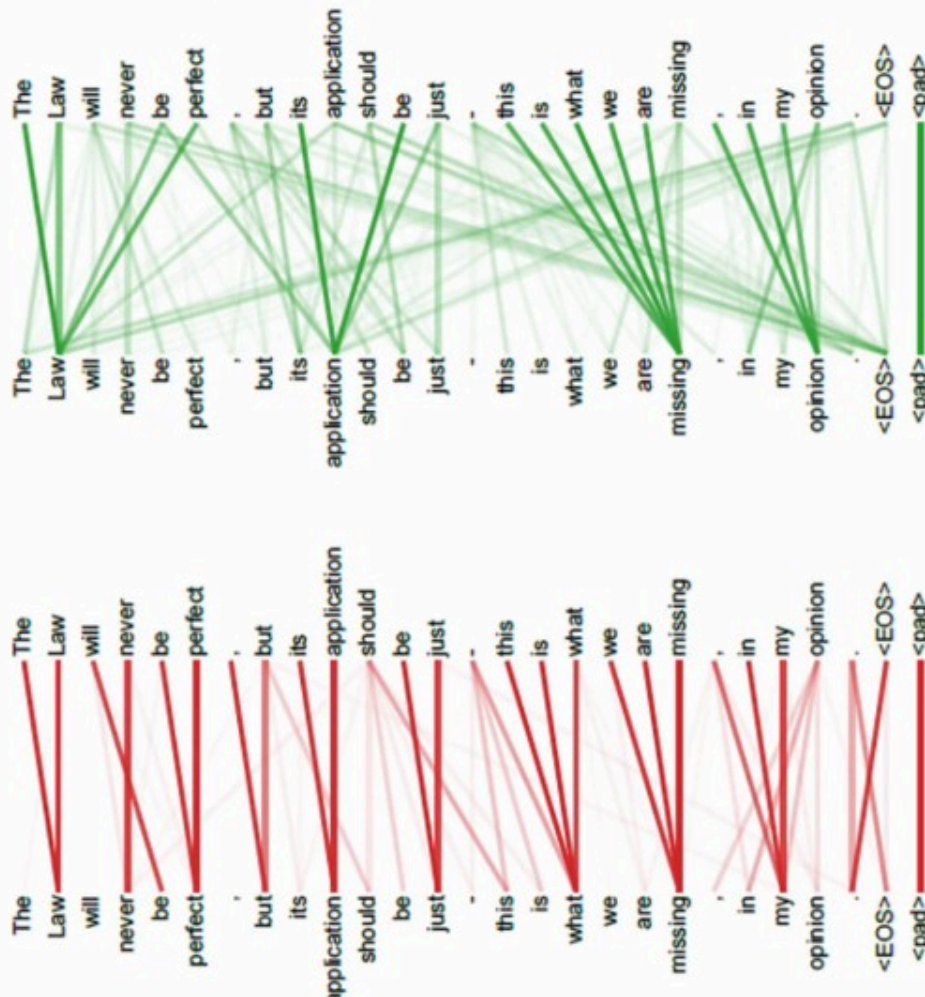
Multi-head Attention



Multi-head Attention Implementation

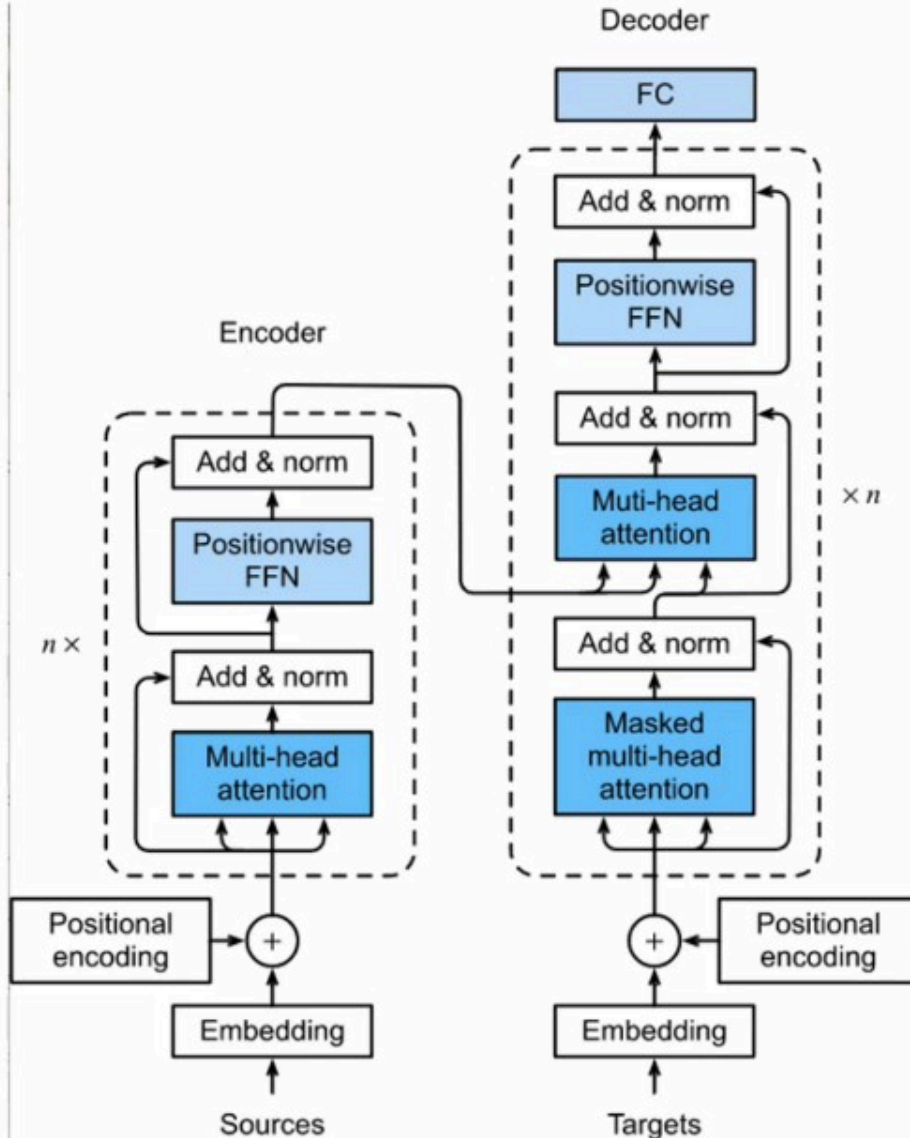


Functions of Multi-Attention Heads



Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L. and Polosukhin, I., 2017. Attention Is All You Need.

Transformer Architecture



Encoder

- The transformer encoder is a stack of multiple identical layers,
- Each layer has two sublayers (either is denoted as).
 - A multi-head self-attention pooling
 - A position-wise feed-forward network (FFN).
- Queries, keys, and values are all from the outputs of the previous encoder layer.
- **Residual Connection:** for any input $\mathbf{x}$ at any position of the sequence, residual connection $\mathbf{x} + \text{residue}(\mathbf{x})$ is computed and followed by layer normalization
- The transformer encoder outputs a d dimensional vector representation for each position of the input sequence

Positional Encoding

- Scaled dot-product and fully-connected layer are permutation invariant
- Sinusoidal positional encoding is a vector of small values (constants) added to the embeddings
- As a result, same word will have slightly different embeddings depending on where they occur in the sentence

$$PE_{pos,2i} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

$$PE_{pos,2i+1} = \cos\left(\frac{pos}{10000^{(2i+1)/d_{model}}}\right)$$

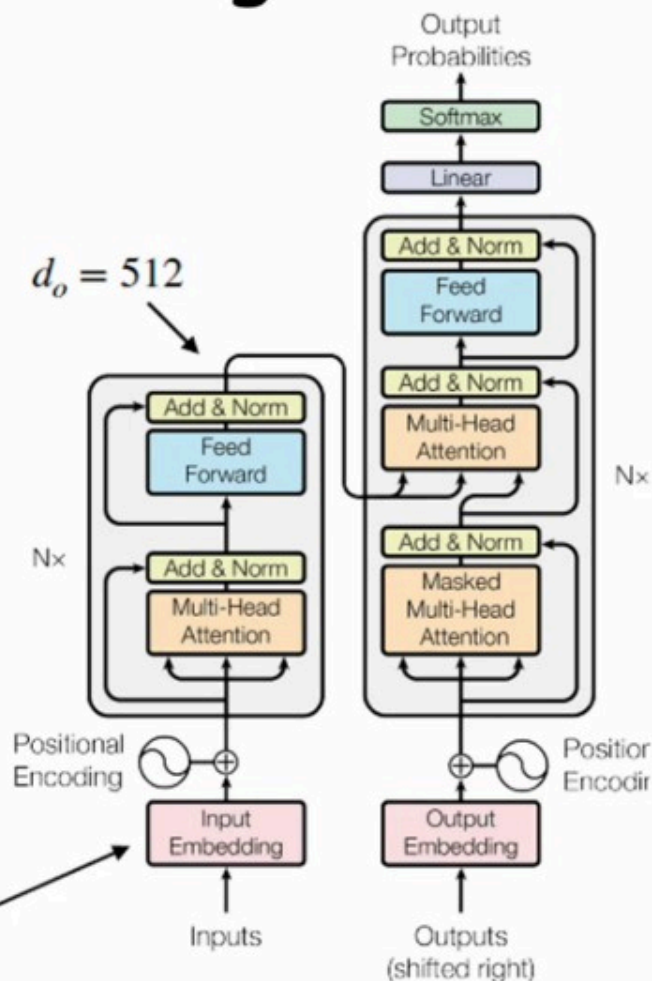
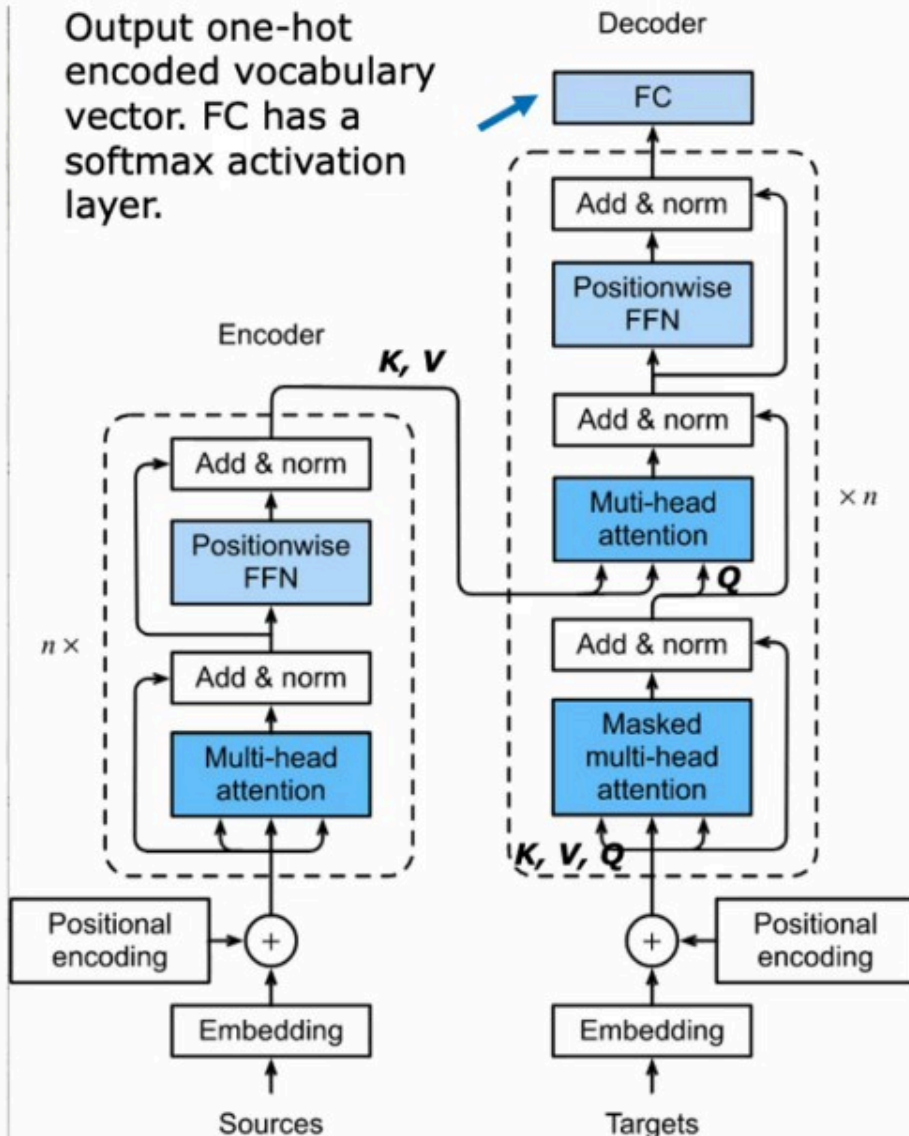


Figure 1: The Transformer - model architecture.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L. and Polosukhin, I., 2017. Attention Is All You Need.

Transformer Architecture

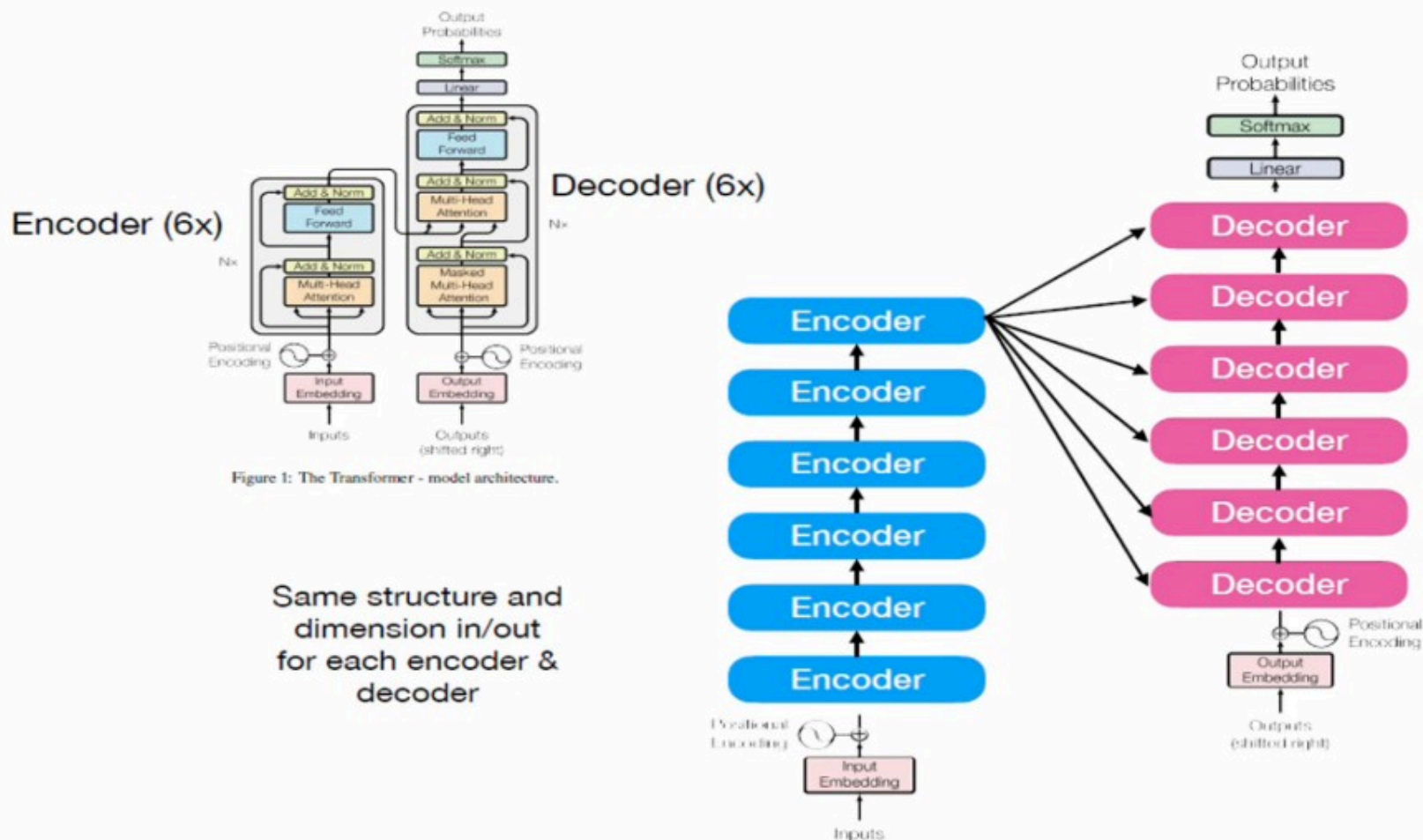
Output one-hot encoded vocabulary vector. FC has a softmax activation layer.



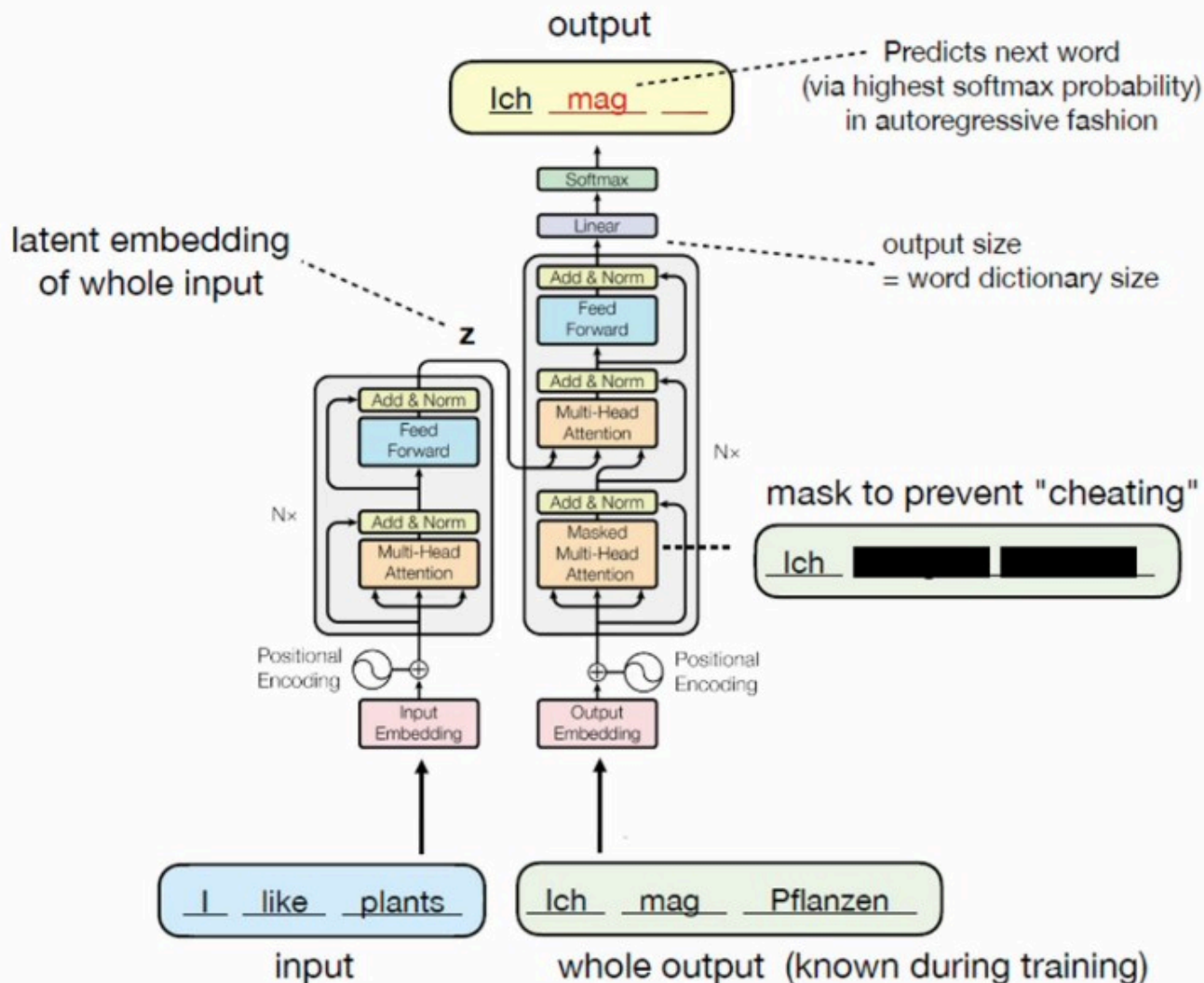
Decoder

- An encoder-decoder attention (EDA) sub-layer is inserted between the two sublayers identical to that in the encoder.
- Inputs to EDA: Key-value pairs are from the encoder, and query is from the previous sub-layer in the decoder.
- each position in the decoder is allowed to only attend to all positions in the decoder up to that position. This masked attention preserves the auto-regressive property, ensuring that the prediction only depends on those output tokens that have been generated

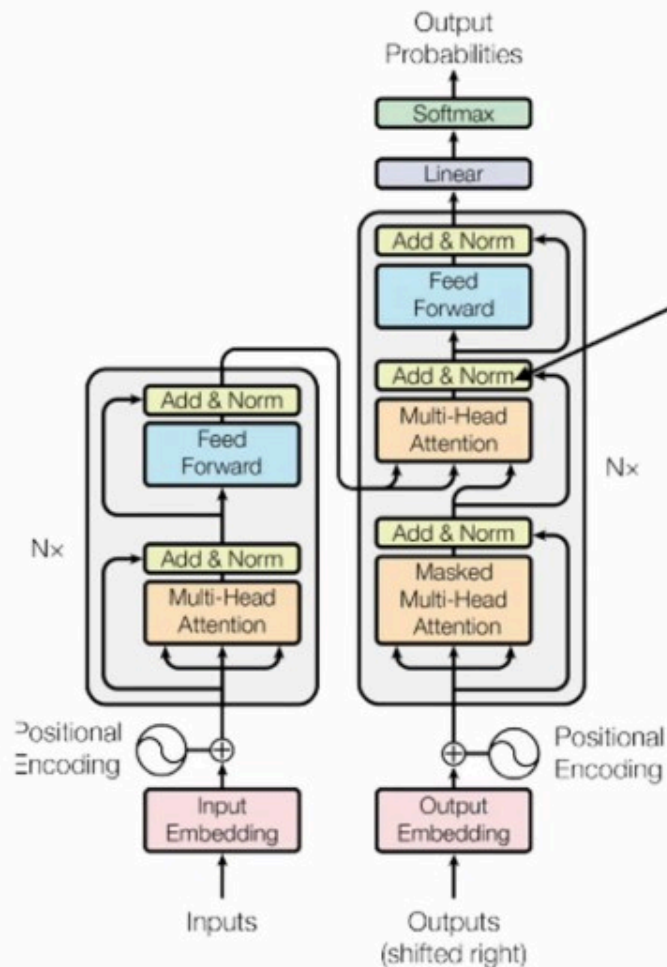
From Encoder to Decoder



Masked Multi-head Attention



Layer Normalization



Layer Normalization

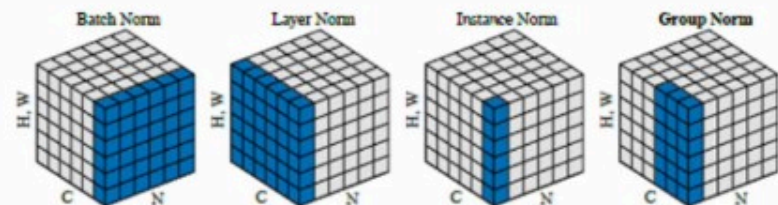


Image source: Wu, Y., & He, K. (2018). Group normalization. ECCV